CSE710
Advanced AI

Lecture 7
Fuzzy TOPSIS
(Technique of Order Preference Similarity
to
the Ideal Solution)

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What is Fuzzy Set?

- Fuzzy sets were introduced independently by Lotfi A. Zadeh and Dieter Klaua in 1965 as an extension of the classical notion of set.

- In mathematics, fuzzy sets are somewhat like sets whose elements have degrees of membership.

What is Fuzzy Set?

Definition: If X is a collection of objects denoted generically by x, then a fuzzy set Ã in X is a set of ordered pairs:

$$\tilde{A} = \{(x, \mu_{\tilde{A}}(x) | x \in X)\}$$

$\mu_{\tilde{A}}(x)$ is called the membership function (generalized characteristic function) which maps X to the membership space M. Its range is the subset of nonnegative real numbers whose supremum (least upper bound) is finite, generally [0,1]

Fuzzy Set Theory

- **Fuzzy** sets **theory** proposes to deal with unclear boundaries, representing vague concepts and working with linguistic variables.

- In this sense, fuzzy sets emerged as an alternative way to deal with uncertainties.

Fuzzy Set Theory

A realtor wants to classify the houses he offers to his clients. One indicator of the comfort of a house is the number of bedrooms in it. Let $X = \{1,2,3,4,\ldots,8\}$ be the set of available types of houses described by x = number of bedrooms in a house. Then the fuzzy set "comfortable type of house for a 4-person family" may be described as

$$\tilde{A} = \{(1,.2),(2,.5),(3,.8),(4,1),(5,.7),(6,.3)\}$$

A fuzzy set is obviously a generalization of a classical set and the membership function is a generalization of the characteristic function. Since we are generally referring to a universal (crisp) set X, some elements of a fuzzy set may have the degree of membership zero. Often it is appropriate to consider those elements of the universe that have a nonzero degree of membership in a fuzzy set:

Fuzzy Set Theory

The *support* of a fuzzy set $\tilde{A}$, $S(\tilde{A})$, is the crisp set of all $x \in X$ such that $\mu_{\tilde{A}}(x) > 0$.

A more general and even more useful notion is that of an α-level set:

The (crisp) set of elements that belong to the fuzzy set $\tilde{A}$ at least to the degree α is called the *α-level set*:

$$A_\alpha = \{x \in X \mid \mu_{\tilde{A}}(x) \geq \alpha\}$$

$$A'_\alpha = \{x \in X \mid \mu_{\tilde{A}}(x) > \alpha\}$$ is called "strong α-level set" or "strong α-cut".

Fuzzy Set Theory

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$$A_{.2} = \{1,2,3,4,5,6\}$$

$$A_{.5} = \{2,3,4,5\}$$

$$A_{.8} = \{3,4\}$$

$$A_1 = \{4\}$$

The strong α-level set for $\alpha = .8$ is $A'_{.8} = \{4\}$

Fuzzy Set Theory

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Fuzzy Set Theory

For a finite fuzzy set $\tilde{A}$ the *cardinality* $|A'|$ is defined as

$$|\tilde{A}| = \sum_{x \in X} \mu_{\tilde{A}}(x)$$

$$\|\tilde{A}\| = \frac{|\tilde{A}|}{|X|}$$ is called the relative cardinality of $\tilde{A}$.

Fuzzy Set Operations

The membership function $\mu_{\tilde{C}}$ of the *intersection* $\tilde{C} = \tilde{A} \cap \tilde{B}$ is pointwise defined for all $x \in X$ by

$$\mu_{\tilde{C}}(x) = \min\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\}$$

The membership function $\mu_{\tilde{D}}$ of the *union* $\tilde{D} = \tilde{A} \cup \tilde{B}$ is pointwise defined for all $x \in X$ by

$$\mu_{\tilde{D}}(x) = \max\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\}$$

The membership function of the *complement* of a fuzzy set $\tilde{A}$, $\mu_{\complement\tilde{A}}$, is defined by

$$\mu_{\complement\tilde{A}}(x) = 1 - \mu_{\tilde{A}}(x) \qquad x \in X$$

$\tilde{A} = \{(1,.2),(2,.5),(3,.8),(4,1),(5,.7),(6,.3)\}$

Fuzzy Set Operations

Let A be the fuzzy set "comfortable type of house for a 4-person family" from example 1.1 and $\tilde{B}$ be the fuzzy set "large type of house" defined as

$$\tilde{B} = \{(3,.2),(4,.4),(5,.6),(6,.8),(7,1),(8,1)\}$$

The intersection $\tilde{C} = \tilde{A} \cap \tilde{B}$ is then

$$\tilde{C} = \{(3,.2),(4,.4),(5,.6),(6,.3)\}$$

The union $\tilde{D} = \tilde{A} \cup \tilde{B}$ is

$$\tilde{D} = \{(1,.2),(2,.5),(3,.8),(4,1),(5,.7),(6,.8),(7,1),(8,1)\}$$

The complement $\complement\tilde{B}$, which might be interpreted as "not large type of house," is

$$\complement\tilde{B} = \{(1,1),(2,1),(3,.8),(4,.6),(5,.4),(6,.2)\}$$

Fuzzy Set Operations

The membership function $\mu_{\tilde{C}}$ of the *intersection* $\tilde{C} = \tilde{A} \cap \tilde{B}$ is pointwise defined for all $x \in X$ by

$$\mu_{\tilde{C}}(x) = \min\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\}$$

The membership function $\mu_{\tilde{D}}$ of the *union* $\tilde{D} = \tilde{A} \cup \tilde{B}$ is pointwise defined for all $x \in X$ by

$$\mu_{\tilde{D}}(x) = \max\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\}$$

The membership function of the *complement* of a fuzzy set $\tilde{A}$, $\mu_{\complement\tilde{A}}$, is defined by

$$\mu_{\complement\tilde{A}}(x) = 1 - \mu_{\tilde{A}}(x) \qquad x \in X$$

Fuzzy Linguistics

Fuzzy logic is primarily associated with quantifying and reasoning out imprecise or vague terms that appear in our languages. These terms are referred to as **linguistic** or **fuzzy variables**.

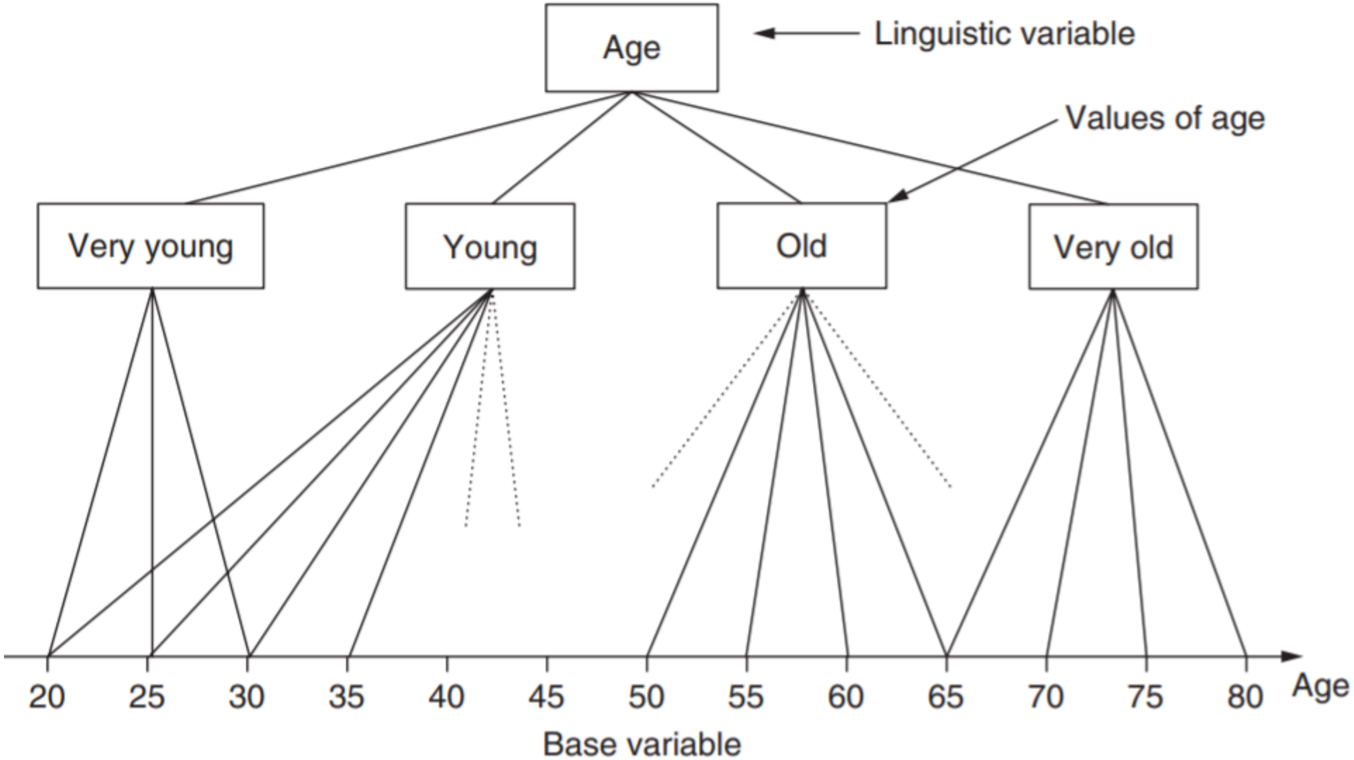
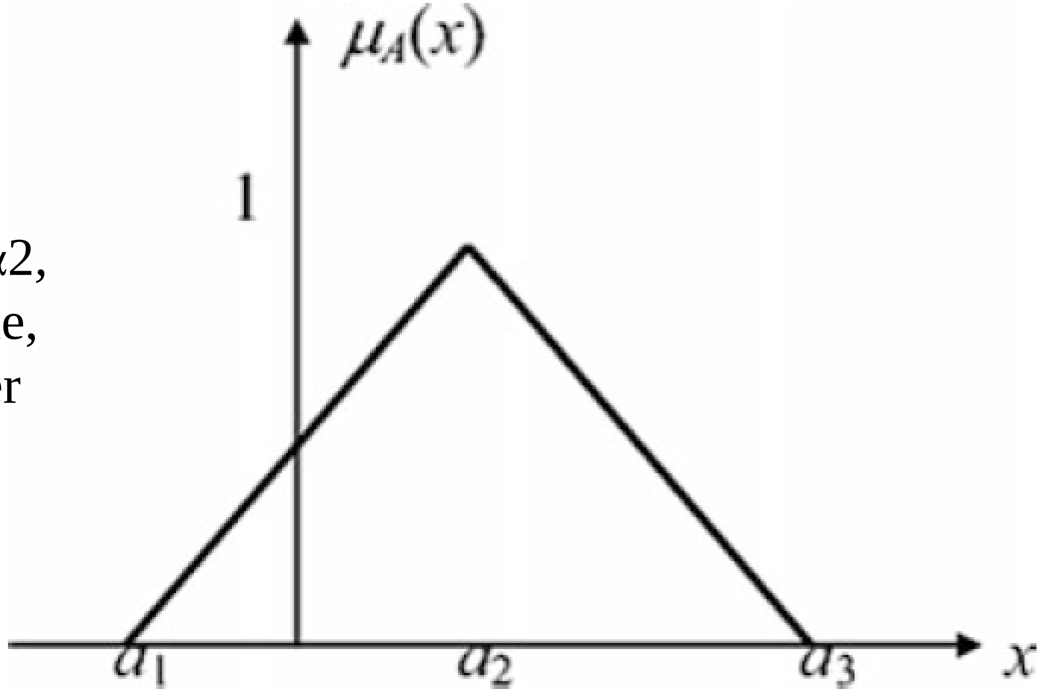


FIGURE 1 | Linguistic variable 'Age'

Triangular Fuzzy Numbers

- One of the most popular shapes of fuzzy numbers is the triangular fuzzy number that can be defined as a triplet $A = (\alpha 1, \alpha 2, \alpha 3)$, shown in the Figure.

A TFN is usually denoted as $A = (\alpha 1, \alpha 2, \alpha 3)$, where $\alpha 2$ indicates the center value, and $\alpha 1$, $\alpha 3$ indicate the lower and upper limit values, respectively



Triangular Fuzzy Numbers

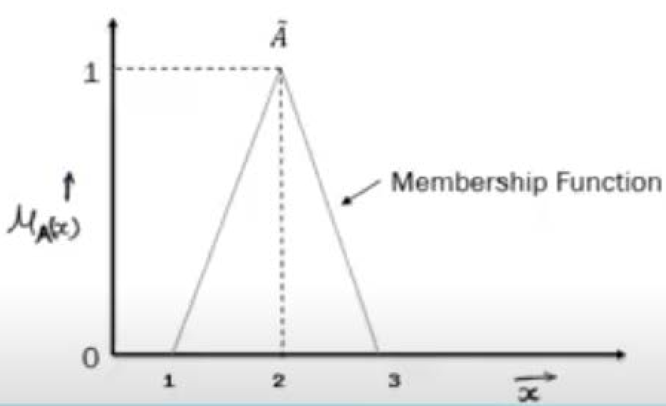
The membership functions are as follows:

$$\mu_A(x) = \begin{cases} 0, & x < \alpha_1 \\ \frac{x-\alpha_1}{\alpha_2-\alpha_1}, & \alpha_1 \le x \le \alpha_2 \\ \frac{\alpha_3-x}{\alpha_3-\alpha_2}, & \alpha_2 \le x \le \alpha_3 \\ 0, & x > \alpha_3 \end{cases}$$

Fuzzification

$$\mu_{\tilde{A}}(x) = \tilde{A} = (1, 2, 3)$$

Fuzzy Number

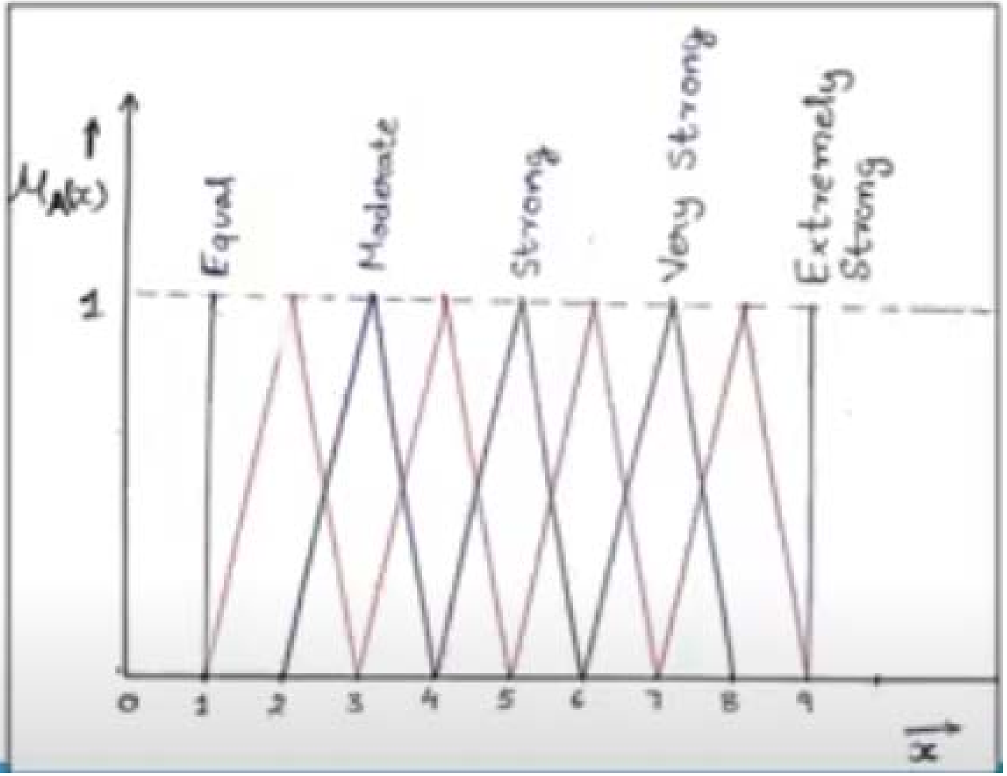


TFN

Fuzzification

Fuzzy Scale of relative importance

Equal	1
Moderate	3
Strong	5
Very strong	7
Extremely strong	9
Intermediate values	2, 4, 6, 8

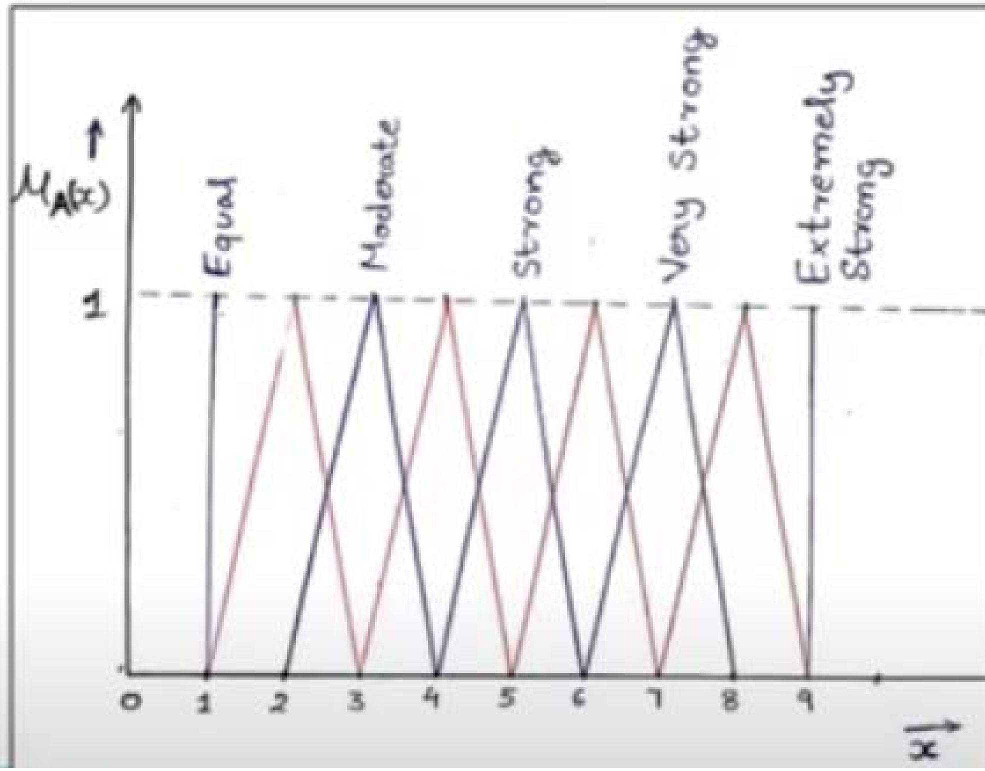


TFN

Fuzzification

Fuzzy Scale of relative importance

Equal	(1, 1, 1)
Moderate	(2, 3, 4)
Strong	(4, 5, 6)
Very strong	(6, 7, 8)
Extremely strong	(9, 9, 9)
Intermediate values	(1, 2, 3)
	(3, 4, 5)
	(5, 6, 7)
	(7, 8, 9)



Fuzzy TOPSIS

- Which one is the best location for …?

A firm is trying to identify the best site out of six possible choices in order to locate a production facility, taking at the same time into account a number of criteria, namely the investment costs, the employment needs, the social impact, and the environmental impact.

The first two criteria are quantitative variables with deterministic values, while the other two are qualitative and rated in a typical Likert scale, ranging from 1 (worst) to 7 (best); all the criteria need to be maximized, since we consider the firm to be a public one.

Fuzzy TOPSIS

- Which one is the best location for …?

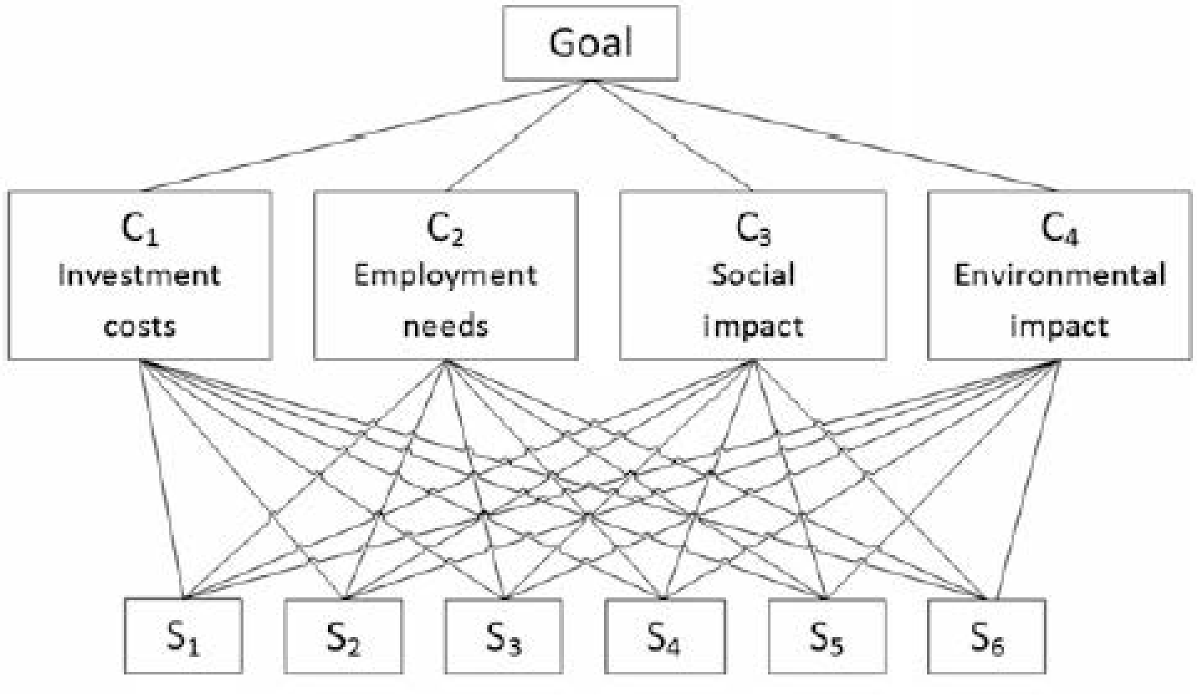


Fig. 1.1 The problem hierarchical structure

Fuzzy TOPSIS

Step 1. Form a Committee of Decision Makers, Then Identify the Evaluation Criteria

If we assume that the decision group has K persons, then the importance of the criteria and the ratings of the alternatives can be calculated as

$$\tilde{x}_{ij} = \frac{1}{K}\left[\tilde{x}_{ij}^{1}(+)\tilde{x}_{ij}^{2}(+)\cdots(+)\tilde{x}_{ij}^{K}\right] \tag{1.19}$$

$$\tilde{w}_{j} = \frac{1}{K}\left[\tilde{w}_{j}^{1}(+)\tilde{w}_{j}^{2}(+)\cdots(+)\tilde{w}_{j}^{K}\right] \tag{1.20}$$

where $\tilde{x}_{ij}^{K}$ and $\tilde{w}_{j}^{K}$ are the ratings and criteria weights of the Kth decision maker.

Fuzzy TOPSIS

Step 2. Choose the Linguistic Variables

Choose the appropriate linguistic variables for the importance weight of the criteria and the linguistic ratings for alternatives with respect to the criteria.

Step 3. Perform Aggregations

Aggregate the weight of criteria to get the aggregated fuzzy weight $\widetilde{w}_j$ of criterion C_j, and pool the decision makers' opinions to get the aggregated fuzzy rating $\widetilde{x}_{ij}$ of alternative A_i under criterion C_j.

Fuzzy TOPSIS

Step 4. Construct the Fuzzy Decision Matrix and the Normalized Fuzzy Decision Matrix

The fuzzy decision matrix is constructed as

$$\widetilde{D} = \begin{bmatrix} \widetilde{x}_{11} & \widetilde{x}_{12} & \cdots & \widetilde{x}_{1n} \\ \widetilde{x}_{21} & \widetilde{x}_{22} & \cdots & \widetilde{x}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \widetilde{x}_{m1} & \widetilde{x}_{m2} & \cdots & \widetilde{x}_{mn} \end{bmatrix} \tag{1.21}$$

and the vector of the criteria weights as

$$\widetilde{W} = [\widetilde{w}_1, \widetilde{w}_2, \cdots, \widetilde{w}_n] \tag{1.22}$$

where $\widetilde{x}_{ij}$ and $\widetilde{w}_j, i = 1, 2, \cdots, m, j = 1, 2, \cdots, n$ are linguistic variables according to Step 2. They can be described by the triangular fuzzy numbers $\widetilde{x}_{ij} = (a_{ij}, b_{ij}, c_{ij})$ and $\widetilde{w}_j = (w_{j1}, w_{j2}, w_{j3})$

Fuzzy TOPSIS

- For the normalization step, Chen uses the linear scale transformation in order to drop the units and make the criteria comparable; it is also important to preserve the property stating that the ranges of the normalized triangular fuzzy numbers belong to [0, 1]. The normalized fuzzy decision matrix denoted by $\tilde{R}$ is:

$$\widetilde{R} = \left[\widetilde{r}_{ij}\right]_{m \times n}$$

Fuzzy TOPSIS

$$\widetilde{R} = [\widetilde{r}_{ij}]_{m \times n} \tag{1.23}$$

The set of benefit criteria is B and the set of cost criteria is C, therefore

$$\widetilde{r}_{ij} = \left(\frac{a_{ij}}{c_j^*}, \frac{b_{ij}}{c_j^*}, \frac{c_{ij}}{c_j^*} \right), \quad j \in B, \quad i = 1, 2, \cdots, m \tag{1.24}$$

$$\widetilde{r}_{ij} = \left(\frac{a_j^-}{c_{ij}}, \frac{a_j^-}{b_{ij}}, \frac{a_j^-}{a_{ij}} \right), \quad j \in C, \quad i = 1, 2, \cdots, m \tag{1.25}$$

$$c_j^* = \max_i c_{ij}, \text{ if } j \in B, \quad i = 1, 2, \cdots, m \tag{1.26}$$

$$a_j^- = \min_i a_{ij}, \text{ if } j \in C, \quad i = 1, 2, \cdots, m \tag{1.27}$$

Fuzzy TOPSIS

Step 5. Construct the Fuzzy Weighted Normalized Decision Matrix
Then, the fuzzy weighted normalized decision matrix can be constructed as

$$\widetilde{V} = [\widetilde{v}_{ij}]_{m \times n}, \quad i = 1, 2, \cdots, m, \quad j = 1, 2, \cdots, n \tag{1.28}$$

where

$$\widetilde{v}_{ij} = \widetilde{r}_{ij}(\cdot)\widetilde{w}_j, \quad i = 1, 2, \cdots, m, \quad j = 1, 2, \cdots, n \tag{1.29}$$

The elements $\widetilde{v}_{ij}, i = 1, 2, \cdots, m, j = 1, 2, \cdots, n$, are normalized positive triangular fuzzy numbers ranging from 0 to 1.

Fuzzy TOPSIS

Step 6. Determine the Fuzzy Positive Ideal Solution (FPIS) and the Fuzzy Negative Ideal Solution (FNIS)

The fuzzy positive ideal solution (FPIS, A^*) and the fuzzy negative ideal solution (FNIS, A^-) are

$$A^* = \left(\widetilde{v}_1^*, \widetilde{v}_2^*, \cdots, \widetilde{v}_n^*\right) \tag{1.30}$$

$$A^- = \left(\widetilde{v}_1^-, \widetilde{v}_2^-, \cdots, \widetilde{v}_n^-\right) \tag{1.31}$$

where

$$\widetilde{v}_j^* = (1, 1, 1), \widetilde{v}_j^- = (0, 0, 0), \quad j = 1, 2, \cdots, n \tag{1.32}$$

Fuzzy TOPSIS

Step 7. Calculate the Distance of Each Alternative from FPIS and FNIS
The distance of each of the alternatives from FPIS and FNIS can be calculated as

$$D_i^* = \sum_{j=1}^{n} d\left(\widetilde{v}_{ij}, \widetilde{v}_j^*\right), \quad i = 1, 2, \cdots, m, \quad j = 1, 2, \cdots, n \tag{1.33}$$

$$D_i^- = \sum_{j=1}^{n} d\left(\widetilde{v}_{ij}, \widetilde{v}_j^-\right), \quad i = 1, 2, \cdots, m, \quad j = 1, 2, \cdots, n \tag{1.34}$$

where d is the distance measurement between two fuzzy numbers

Chen [5] uses the vertex method to calculate the distance between two triangular fuzzy numbers $\widetilde{m} = (m_1, m_2, m_3)$ and $\widetilde{n} = (n_1, n_2, n_3)$ as

$$d(\widetilde{m}, \widetilde{n}) = \sqrt{\frac{1}{3}\left[(m_1 - n_1)^2 + (m_2 - n_2)^2 + (m_3 - n_3)^2\right]} \tag{1.17}$$

Fuzzy TOPSIS

Step 8. Calculate the Closeness Coefficient of Each Alternative

The closeness coefficient of each alternative can be defined as

$$CC_i = \frac{D_i^-}{D_i^* + D_i^-}, \quad i = 1, 2, \cdots, m \tag{1.35}$$

Step 9. Rank the Alternatives

An alternative A_i is better than A_j if its closeness coefficient is closer to 1.

Therefore, the final ranking of the alternatives is defined by the value of the closeness coefficient.

Example

	Attribute Or Criteria	Communication Skills	Knowledge	Time To Solve Problem
Alternative	Candidate-1	Average	Very High	High
	Candidate-2	High	Very High	Average
	Candidate-3	Very high	Average	Low
	Candidate-4	Low	Average	Low

Example

Attribute Or Criteria	Communication Skills	Knowledge	Time To Solve Problem
Candidate-1	Average	Very High	High
Candidate-2	High	Very High	Average
Candidate-3	Very high	Average	Low
Candidate-4	Low	Average	Low

Alternative

Term	Fuzzy Number
Very Low	1,1,3
Low	1,3,5
Average	3,5,7
High	5,7,9
Very High	7,9,9

Example

Decision Maker-1

Attribute Or Criteria	Communication Skills	Knowledge	Time To Solve Problem
Candidate-1	Average	Very High	High
Candidate-2	High	Very High	Average
Candidate-3	Very high	Average	Low
Candidate-4	Low	Average	Very Low

Decision Maker-2

Attribute Or Criteria	Communication Skills	Knowledge	Time To Solve Problem
Candidate-1	High	Very High	High
Candidate-2	High	High	Average
Candidate-3	Very high	Average	Very Low
Candidate-4	Low	Average	Very Low

Decision Maker-3

Attribute Or Criteria	Communication Skills	Knowledge	Time To Solve Problem
Candidate-1	Average	High	High
Candidate-2	High	Average	Average
Candidate-3	High	Average	Low
Candidate-4	Very Low	Low	Very Low

Example

Decision Maker-1=k

Attribute Or Criteria	Comm. Skills	Knowledge	Time To Solve Problem
C-1	3,5,7	7,9,9	5,7,9
C-2	5,7,9	7,9,9	3,5,7
C-3	7,9,9	[illegible]	[illegible]
C-4	1,3,5	3,5,7	1,1,3

$a_{11}^{1}, b_{11}^{1}, c_{11}^{1}$

Decision Maker-2=k

Attribute Or Criteria	Comm. Skills	Knowledge	Time To Solve Problem
C-1	5,7,9	7,9,9	5,7,9
C-2	5,7,9	5,7,9	3,5,7
C-3	7,9,9	[illegible]	[illegible]
C-4	1,3,5	3,5,7	1,1,3

$a_{11}^{2}, b_{11}^{2}, c_{11}^{2}$

Decision Maker-3=k

Attribute Or Criteria	Comm. Skills	Knowledge	Time To Solve Problem
C-1	3,5,7	5,7,9	5,7,9
C-2	5,7,9	3,5,7	3,5,7
C-3	5,7,9	[illegible]	[illegible]
C-4	1,1,3	1,3,5	1,1,3

$a_{11}^{3}, b_{11}^{3}, c_{11}^{3}$

Co

a_{11} =min(3,5,3)=3

Attribute Or Criteria	Comm. Skills	Knowledge	Time To Solve Problem
C-1	3,	$\widetilde{X}_{12}$	$\widetilde{X}_{13}$
C-2	$\widetilde{X}_{21}$	$\widetilde{X}_{22}$	$\widetilde{X}_{23}$
C-3	$\widetilde{X}_{31}$	$\widetilde{X}_{32}$	$\widetilde{X}_{33}$
C-4	$\widetilde{X}_{41}$	$\widetilde{X}_{42}$	$\widetilde{X}_{43}$

$$\tilde{x}_{ij} = (a_{ij}, b_{ij}, c_{ij})$$

$$a_{ij} = \min_k \{a_{ij}^k\}\ ,\ b_{ij} = \frac{1}{K}\sum_{k=1}^{K} b_{ij}^k\ ,\ c_{ij} = \max_k \{c_{ij}^k\}$$

Example

Decision Maker-1

Attribute Or Criteria	Comm. Skills	Knowledge	Time To Solve Problem
C-1	3,5,7	7,9,9	5,7,9
C-2	5,7,9	7,9,9	3,5,7
C-3	7,9,9	3,5,7	1,3,5
C-4	1,3,5	3,5,7	1,1,3

Decision Maker-2

Attribute Or Criteria	Comm. Skills	Knowledge	Time To Solve Problem
C-1	5,7,9	7,9,9	5,7,9
C-2	5,7,9	5,7,9	3,5,7
C-3	7,9,9	3,5,7	1,1,3
C-4	1,3,5	3,5,7	1,1,3

Decision Maker-3

Attribute Or Criteria	Comm. Skills	Knowledge	Time To Solve Problem
C-1	3,5,7	5,7,9	5,7,9
C-2	5,7,9	3,5,7	3,5,7
C-3	5,7,9	3,5,7	1,3,5
C-4	1,1,3	1,3,5	1,1,3

Combined Decision Matrix

Attribute Or Criteria	Comm. Skills	Knowledge	Time To Solve Problem
C-1	3, 5.667 , 9	5, 8.333 ,9	5 ,7 ,9
C-2	5 , 7 , 9	3 ,7 ,9	3 , 5 , 7
C-3	5 , 8.333 ,9	3 , 5 ,7	1 , 2.333 , 5
C-4	1 , 2.333 ,5	1 , 4.333 ,7	1 , 1 , 3

$$\tilde{x}_{ij} = (a_{ij}, b_{ij}, c_{ij})$$

$$a_{ij} = \min_k\{a^k_{ij}\}\ ,\ b_{ij} = \frac{1}{K}\sum_{k=1}^{K} b^k_{ij}\ ,\ c_{ij} = \max_k\{c^k_{ij}\}$$

Manoj Mathew

Example

Weightage →	5, 7, 9	7, 9, 9	3, 5, 7
Attribute Or Criteria ⇒	**Communication Skills**	**Knowledge**	**Time To Solve Problem**
Alternative — Candidate-1	3, 5.667 , 9	5, 8.333 ,9	5 ,7 ,9
Candidate-2	5 , 7 , 9	3 ,7 ,9	3 , 5 , 7
Candidate-3	5 , 8.333 ,9	3 , 5 ,7	1 , 2.333 , 5
Candidate-4	1 , 2.333 ,5	1 , 4.333 ,7	1 , 1 , 3

$$w_{j1} = \min_k \{w_{j1}^k\}, \quad w_{j2} = \frac{1}{K}\sum_{k=1}^{K} w_{j2}^k, \quad w_{j3} = \max_k \{w_{j3}^k\}$$

$$a_{ij} = \min_k \{a_{ij}^k\}, \quad b_{ij} = \frac{1}{K}\sum_{k=1}^{K} b_{ij}^k, \quad c_{ij} = \max_k \{c_{ij}^k\}$$

Example

Weightage →	5, 7, 9	7, 9, 9	3, 5, 7
Attribute Or Criteria →	**Communication Skills**	**Knowledge**	**Time To Solve Problem**
Candidate-1	$\frac{3}{9}, \frac{5.667}{9}, \frac{9}{9}$	$\frac{5}{9}, \frac{8.333}{9}, \frac{9}{9}$	$\frac{1}{9}, \frac{1}{7}, \frac{1}{5}$
Candidate-2	$\frac{5}{9}, \frac{7}{9}, \frac{9}{9}$	$\frac{3}{9}, \frac{7}{9}, \frac{9}{9}$	$\frac{1}{7}, \frac{1}{5}, \frac{1}{3}$
Candidate-3	$\frac{5}{9}, \frac{8.333}{9}, \frac{9}{9}$	$\frac{3}{9}, \frac{5}{9}, \frac{7}{9}$	$\frac{1}{5}, \frac{1}{2.333}, \frac{1}{1}$
Candidate-4	$\frac{1}{9}, \frac{2.333}{9}, \frac{9}{9}$	$\frac{3}{9}, \frac{5.667}{9}, \frac{9}{9}$	$\frac{1}{3}, \frac{1}{1}, \frac{1}{1}$

Alternative

Compute the normalized fuzzy decision matrix

$$\tilde{r}_{ij} = \left(\frac{a_{ij}}{c_j^*}, \frac{b_{ij}}{c_j^*}, \frac{c_{ij}}{c_j^*}\right) \text{ and } c_j^* = \max_i\{c_{ij}\} \text{ (benefit criteria)}$$

$$\tilde{r}_{ij} = \left(\frac{a_j^-}{c_{ij}}, \frac{a_j^-}{b_{ij}}, \frac{a_j^-}{a_{ij}}\right) \text{ and } a_j^- = \min_i\{a_{ij}\} \text{ (cost criteria)}$$

Example

Weightage	5, 7, 9	7, 9, 9	3, 5, 7
Attribute Or Criteria	Communication Skills	Knowledge	Time To Solve Problem
Candidate-1	0.333 , 0.630 , 1	0.556 , 0.926 , 1	0.111 , 0.143 , 0.2
Candidate-2	0.556 , 0.778 , 1	0.333 , 0.778 , 1	0.143 , 0.2 , 0.333
Candidate-3	0.556 , 0.926 , 1	0.333 , 0.556 , 0.778	0.2 , 0.429 , 1
Candidate-4	0.111 , 0.259 , 0.556	0.111 , 0.481 , 0.778	0.333 , 1 , 1

Alternative

Compute the weighted normalized fuzzy decision matrix

Manoj Mathew

$$\tilde{v}_{ij} = \tilde{r}_{ij} \times w_j$$

$$\widetilde{A_1} \otimes \widetilde{A_2} = (a_1, b_1, c_1) \otimes (a_2, b_2, c_2) = (a_1 * a_2, b_1 * b_2, c_1 * c_2)$$

Example

	Weightage	5, 7, 9	7, 9, 9	3, 5, 7
	Attribute Or Criteria	**Communication Skills**	**Knowledge**	**Time To Solve Problem**
Alternative	Candidate-1	1.667 , 4.407 , 9	3.889 , 8.333 , 9	0.333 , 0.714 , 1.4
	Candidate-2	2.778 , 5.444 , 9	2.333 , 7 , 9	0.429 , 1 , 2.333
	Candidate-3	2.778 , 6.481 , 9	2.333 , 5 , 7	0.6 , 2.143 , 7
	Candidate-4	0.556 ,1.815 , 5	0.778 , 4.333 , 7	1 , 5 , 7

Compute the weighted normalized fuzzy decision matrix

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$$\tilde{v}_{ij} = \tilde{r}_{ij} \times w_j$$

Example

Alternative

Attribute Or Criteria	Communication Skills	Knowledge	Time To Solve Problem
Candidate-1	1.667 , 4.407 , 9	3.889 , 8.333 , 9	0.333 , 0.714 , 1.4
Candidate-2	2.778 , 5.444 , 9	2.333 , 7 , 9	0.429 , 1 , 2.333
Candidate-3	2.778 , 6.481 , 9	2.333 , 5 , 7	0.6 , 2.143 , 7
Candidate-4	0.556 , 1.815 , 5	0.778 , 4.333 , 7	1 , 5 , 7

Compute the Fuzzy Positive Ideal Solution (FPIS)

Fuzzy Negative Ideal Solution (FNIS)

$$A^* = (\tilde{v}_1^*, \tilde{v}_2^*, \cdots, \tilde{v}_n^*), \text{ where } \tilde{v}_j^* = \max_i \{v_{ij3}\}$$

$$A^- = (\tilde{v}_1^-, \tilde{v}_2^-, \cdots, \tilde{v}_n^-), \text{ where } \tilde{v}_j^- = \min_i \{v_{ij1}\}$$

Example

Alternative

Attribute Or Criteria	Communication Skills	Knowledge	Time To Solve Problem
Candidate-1	1.667 , 4.407 , 9	3.889 , 8.333 , 9	0.333 , 0.714 , 1.4
Candidate-2	2.778 , 5.444 , 9	2.333 , 7 , 9	0.429 , 1 , 2.333
Candidate-3	2.778 , 6.481 , 9	2.333 , 5 , 7	0.6 , 2.143 , 7
Candidate-4	0.556 , 1.815 , 5	0.778 , 4.333 , 7	1 , 5 , 7
A^*	2.778 , 6.481 , 9	3.889 , 8.333 , 9	1 , 5 , 7
A^-	0.556 ,1.815 , 5	0.778 , 4.333 , 7	0.333 , 0.714 , 1.4

$$A^* = (\tilde{v}_1^*, \tilde{v}_2^*, \cdots, \tilde{v}_n^*), \text{ where } \tilde{v}_j^* = \max_i \{v_{ij3}\}$$

$$A^- = (\tilde{v}_1^-, \tilde{v}_2^-, \cdots, \tilde{v}_n^-), \text{ where } \tilde{v}_j^- = \min_i \{v_{ij1}\}$$

Example

Attribute Or Criteria	Communication Skills	Knowledge	Time To Solve Problem
Candidate-1	1.667 , 4.407 , 9	3.889 , 8.333 , 9	0.333 , 0.714 , 1.4
Candidate-2	2.778 , 5.444 , 9	2.333 , 7 , 9	0.429 , 1 , 2.333
Candidate-3	2.778 , 6.481 , 9	2.333 , 5 , 7	0.6 , 2.143 , 7
Candidate-4	0.556 ,1.815 , 5	0.778 , 4.333 , 7	1 , 5 , 7
A^*	2.778 , 6.481 , 9	3.889 , 8.333 , 9	1 , 5 , 7

$$d(\tilde{x}, \tilde{y}) = \sqrt{\frac{1}{3}\left[(a_1 - a_2)^2 + (b_1 - b_2)^2 + (c_1 - c_2)^2\right]}$$

Candidate-1	1.358	0.000	4.089
Candidate-2			3.564
Candidate-3			1.666
Candidate-4			0.000

$$\sqrt{\frac{1}{3}\left[(1.667 - 2.778)^2 + (4.407 - 6.841)^2 + (9 - 9)^2\right]}$$

Example

Attribute Or Criteria	Communication Skills	Knowledge	Time To Solve Problem
Candidate-1	1.667 , 4.407 , 9	3.889 , 8.333 , 9	0.333 , 0.714 , 1.4
Candidate-2	2.778 , 5.444 , 9	2.333 , 7 , 9	0.429 , 1 , 2.333
Candidate-3	2.778 , 6.481 , 9	2.333 , 5 , 7	0.6 , 2.143 , 7
Candidate-4	0.556 , 1.815 , 5	0.778 , 4.333 , 7	1 , 5 , 7
A^-	0.556 , 1.815 , 5	0.778 , 4.333 , 7	0.333 , 0.714 , 1.4

$$d(\tilde{x}, \tilde{y}) = \sqrt{\frac{1}{3}\left[(a_1 - a_2)^2 + (b_1 - b_2)^2 + (c_1 - c_2)^2\right]}$$

Candidate-1	2.826	3.145	0.000
Candidate-2	3.372	2.124	0.566
Candidate-3	3.773	0.977	3.340
Candidate-4	0.000	0.000	4.089

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Example

Compute the distance from each alternative to the FPIS and to the FNIS

Fuzzy Positive Ideal Solution

Attribute Or Criteria	Communication Skills	Knowledge	Time To Solve Problem
Candidate-1	1.358	0.000	4.089
Candidate-2	0.599	1.183	3.564
Candidate-3	0.000	2.417	1.666
Candidate-4	3.773	3.145	0.000

$$d_i^* = \sum_{j=1}^{n} d(\tilde{v}_{ij}, \tilde{v}_j^*)$$

Fuzzy Negative Ideal Solution

Attribute Or Criteria	Communication Skills	Knowledge	Time To Solve Problem
Candidate-1	2.826	3.145	0.000
Candidate-2	3.372	2.124	0.566
Candidate-3	3.773	0.977	3.340
Candidate-4	0.000	0.000	4.089

$$d_i^- = \sum_{j=1}^{n} d(\tilde{v}_{ij}, \tilde{v}_j^-)$$

Example

Fuzzy Positive Ideal Solution

Attribute Or Criteria	Communication Skills	Knowledge	Time To Solve Problem	d_i^*
Candidate-1	1.358	0.000	4.089	5.448
Candidate-2	0.599	1.183	3.564	5.346
Candidate-3	0.000	2.417	1.666	4.083
Candidate-4	3.773	3.145	0.000	6.919

Fuzzy Positive Ideal Solution

Attribute Or Criteria	Communication Skills	Knowledge	Time To Solve Problem	d_i^-
Candidate-1	2.826	3.145	0.000	5.971
Candidate-2	3.372	2.124	0.566	6.062
Candidate-3	3.773	0.977	3.340	8.091
Candidate-4	0.000	0.000	4.089	4.089

Example

Compute the closeness coefficient CC_i for each alternative

$$CC_i = \frac{d_i^-}{d_i^- + d_i^*}$$

Attribute Or Criteria	d_i^*	d_i^-
Candidate-1	5.448	5.971
Candidate-2	5.346	6.062
Candidate-3	4.083	8.091
Candidate-4	6.919	4.089

Example

Compute the closeness coefficient CC_i for each alternative

$$CC_i = \frac{d_i^-}{d_i^- + d_i^*}$$

Attribute Or Criteria	d_i^*	d_i^-	CC_i	*Rank*
Candidate-1	5.448	5.971	0.522909	3
Candidate-2	5.346	6.062	0.531406	2
Candidate-3	4.083	8.091	0.664606	1
Candidate-4	6.919	4.089	0.371494	4

Fuzzy TOPSIS

Linguistic variables for the importance weight of each criterion	
Very low (VL)	(0, 0, 0.1)
Low (L)	(0, 0.1, 0.3)
Medium low (ML)	(0.1, 0.3, 0.5)
Medium (M)	(0.3, 0.5, 0.7)
Medium high (MH)	(0.5, 0.7, 0.9)
High (H)	(0.7, 0.9, 1.0)
Very high (VH)	(0.9, 1.0, 1.0)

Table 1.12 Linguistic variables for the criteria

Fuzzy TOPSIS

Linguistic variables for the ratings	
Very poor (VP)	(0, 0, 1)
Poor (P)	(0, 1, 3)
Medium poor (MP)	(1, 3, 5)
Fair (F)	(3, 5, 7)
Medium good (MG)	(5, 7, 9)
Good (G)	(7, 9, 10)
Very good (VG)	(9, 10, 10)

Table 1.13 Linguistic variables for the ratings

Fuzzy TOPSIS

	D_1	D_2	D_3
Investment costs	H	VH	VH
Employment needs	M	H	VH
Social impact	M	MH	ML
Environmental impact	H	VH	MH

Table 1.14 The importance weight of the criteria for each decision maker

Criteria	Candidate sites	Decision makers		
		D_1	D_2	D_3
Investment costs	Site 1	VG	G	MG
	Site 2	MP	F	F
	Site 3	MG	MP	F
	Site 4	MG	VG	VG
	Site 5	VP	P	G
	Site 6	F	G	G
Employment needs	Site 1	F	MG	MG
	Site 2	F	VG	G
	Site 3	MG	MG	VG
	Site 4	G	G	VG
	Site 5	P	VP	MP
	Site 6	F	MP	MG
Social impact	Site 1	P	P	MP
	Site 2	MG	VG	G
	Site 3	MP	F	F
	Site 4	MG	VG	G
	Site 5	G	G	VG
	Site 6	VG	MG	F
Environmental impact	Site 1	G	VG	G
	Site 2	MG	F	MP
	Site 3	MP	P	P
	Site 4	VP	F	P
	Site 5	G	MG	MG
	Site 6	P	MP	F

Table 1.15 The ratings of the six sites by the three decision makers for the four criteria

Criteria	Candidate sites	Decision makers		
		D_1	D_2	D_3
Investment costs	Site 1	9,10,10	7,9,10	5,7,9
	Site 2	MP	F	F
	Site 3	MG	MP	F
	Site 4	MG	VG	VG
	Site 5	VP	P	G
	Site 6	F	G	G
Employment needs	Site 1	3,5,7	5,7,9	5,7,9
	Site 2	F	VG	G
	Site 3	MG	MG	VG
	Site 4	G	G	VG
	Site 5	P	VP	MP
	Site 6	3,5,7	1,3,5	5,7,9
Social impact	Site 1	P	P	MP
	Site 2	MG	VG	G
	Site 3	MP	F	F
	Site 4	MG	VG	G
	Site 5	G	G	VG
	Site 6	VG	MG	F
Environmental impact	Site 1	7,9,10	9,10,10	7,9,10
	Site 2	MG	F	MP
	Site 3	MP	P	P
	Site 4	VP	F	P
	Site 5	G	MG	MG
	Site 6	P	MP	F

Fuzzy TOPSIS

Table 1.16 Fuzzy decision matrix and fuzzy weights

	Investment costs	Employment needs	Social impact	Environmental impact
Site 1	(7.000, 8.667, 9.667)	(4.333, 6.333, 8.333)	(0.333, 1.667, 3.667)	(7.667, 9.333, 10.000)
Site 2	(2.333, 4.333, 6.333)	(6.333, 8.000, 9.000)	(7.000, 8.667, 9.667)	(3.000, 5.000, 7.000)
Site 3	(3.000, 5.000, 7.000)	(6.333, 8.000, 9.333)	(2.333, 4.333, 6.333)	(0.333, 1.667, 3.667)
Site 4	(7.667, 9.000, 9.667)	(7.667, 9.333, 10.000)	(7.000, 8.667, 9.667)	(1.000, 2.000, 3.667)
Site 5	(2.333, 3.333, 4.667)	(0.333, 1.333, 3.000)	(7.667, 9.333, 10.000)	(5.667, 7.667, 9.333)
Site 6	(5.667, 7.667, 9.000)	(3.000, 5.000, 7.000)	(5.667, 7.333, 8.667)	(1.333, 3.000, 5.000)
Weight	(0.833, 0.967, 1.000)	(0.633, 0.800, 0.900)	(0.300, 0.500, 0.700)	(0.700, 0.867, 0.967)

	D_1	D_2	D_3
Investment costs (0.7,0.9.1.0)	H	VH (0.9,1.0,1.0)	VH (0.9,1.0,1.0)
Employment needs (0.3,0.5,0.7)	M	H	VH
Social impact (0.3,0.5,0.7)	M	MH	ML
Environmental impact	H	VH	MH

$$\tilde{w}_j = \frac{1}{K}\left[\tilde{w}_j^1(+)\tilde{w}_j^2(+)\cdots(+)\tilde{w}_j^K\right]$$

w_1=((0.7+0.9+0.9)/3, (0.9+1+1)/3, (1+1+1)/3)
=(0.833, 0.966, 1)

$$\widetilde{x_{1,1}} = \left(\frac{9+7+5}{3} = 7.0, \frac{10+9+7}{3} = 8.6667, \frac{10+10+9}{3} = 9.6667\right)$$

$$\tilde{x}_{ij} = \frac{1}{K}\left[\tilde{x}_{ij}^1(+)\tilde{x}_{ij}^2(+)\cdots(+)\tilde{x}_{ij}^K\right]$$

Fuzzy TOPSIS

Table 1.17 Fuzzy normalized decision matrix

	Investment costs	Employment needs	Social impact	Environmental impact
Site 1	(0.700, 0.867, 0.967)	(0.433, 0.633, 0.833)	(0.033, 0.167, 0.367)	(0.767, 0.933, 1.000)
Site 2	(0.233, 0.433, 0.633)	(0.633, 0.800, 0.900)	(0.700, 0.867, 0.967)	(0.300, 0.500, 0.700)
Site 3	(0.300, 0.500, 0.700)	(0.633, 0.800, 0.933)	(0.233, 0.433, 0.633)	(0.033, 0.167, 0.367)
Site 4	(0.767, 0.900, 0.967)	(0.767, 0.933, 1.000)	(0.700, 0.867, 0.967)	(0.100, 0.200, 0.367)
Site 5	(0.233, 0.333, 0.467)	(0.033, 0.133, 0.300)	(0.767, 0.933, 1.000)	(0.567, 0.767, 0.933)
Site 6	(0.567, 0.767, 0.900)	(0.300, 0.500, 0.700)	(0.567, 0.733, 0.867)	(0.133, 0.300, 0.500)

Fuzzy TOPSIS

Table 1.18 Fuzzy weighted normalized decision matrix

	Investment costs	Employment needs	Social impact	Environmental impact
Site 1	(0.700, 0.867, 0.967)	(0.433, 0.633, 0.833)	(0.033, 0.167, 0.367)	(0.767, 0.933, 1.000)
Site 2	(0.233, 0.433, 0.633)	(0.633, 0.800, 0.900)	(0.700, 0.867, 0.967)	(0.300, 0.500, 0.700)
Site 3	(0.300, 0.500, 0.700)	(0.633, 0.800, 0.933)	(0.233, 0.433, 0.633)	(0.033, 0.167, 0.367)
Site 4	(0.767, 0.900, 0.967)	(0.767, 0.933, 1.000)	(0.700, 0.867, 0.967)	(0.100, 0.200, 0.367)
Site 5	(0.233, 0.333, 0.467)	(0.033, 0.133, 0.300)	(0.767, 0.933, 1.000)	(0.567, 0.767, 0.933)
Site 6	(0.567, 0.767, 0.900)	(0.300, 0.500, 0.700)	(0.567, 0.733, 0.867)	(0.133, 0.300, 0.500)

Fuzzy TOPSIS

Table 1.19 The final result

Alternative	D_i^*	D_i^-	Closeness coefficient	Final ranking
Site 1	1.965	2.305	0.540	1
Site 2	2.212	2.051	0.481	3
Site 3	2.577	1.673	0.394	6
Site 4	1.959	2.279	0.538	2
Site 5	2.526	1.704	0.403	5
Site 6	2.347	1.914	0.449	4

Thank you

Thank You